

MsgGuard: Generative AI for Phishing Detection and Prevention

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What Is Phishing?

- Phishing is one of the most common cyber-attacks today
- Attackers imitate trusted brands (PayPal, Amazon, banks, etc.)
- Users are tricked into clicking malicious links
- Phishing causes financial loss, identity theft, and reputational damage

The Problem We Address

- Modern phishing URLs look very similar to legitimate ones

Attackers use:

- Newly registered domains
- Brand-like domain names
- Legitimate-looking hosting origin (hosting provider, IP address, country/region)
- Many detection systems fail on **new and unseen phishing URLs (zero-day attacks)**



Existing Solutions Today

Truecaller



- Detects spam and phishing messages based on sender reputation and patterns
- Useful for blocking known scam numbers

URLScan.io



- Analyzes the behavior and structure of URLs and websites
- Provides technical indicators (redirects, hosting, scripts)

Norton SMS Protection



- Detects malicious or suspicious SMS content
- Uses predefined patterns and threat intelligence

Lacking Against Modern Threats



Struggle with zero-day attacks



Cannot analyze deceptive language



Weak at detecting new domains and brands

AI-Based Phishing Detection System

We propose an AI-based phishing detection system that:



Analyzes message content to identify impersonation and deceptive language



Verifies link identity instead of relying only on blacklists



Uses Domain age, hosting origin, and brand matching

Enables effective detection of brand impersonation and zero-day phishing attacks

Project goal

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Phishing Risk Reduction

Reduce the impact of phishing attacks delivered through messages and links.



Safer User Decisions

Help users make safer and more informed decisions when receiving suspicious content.



Website Identity Verification

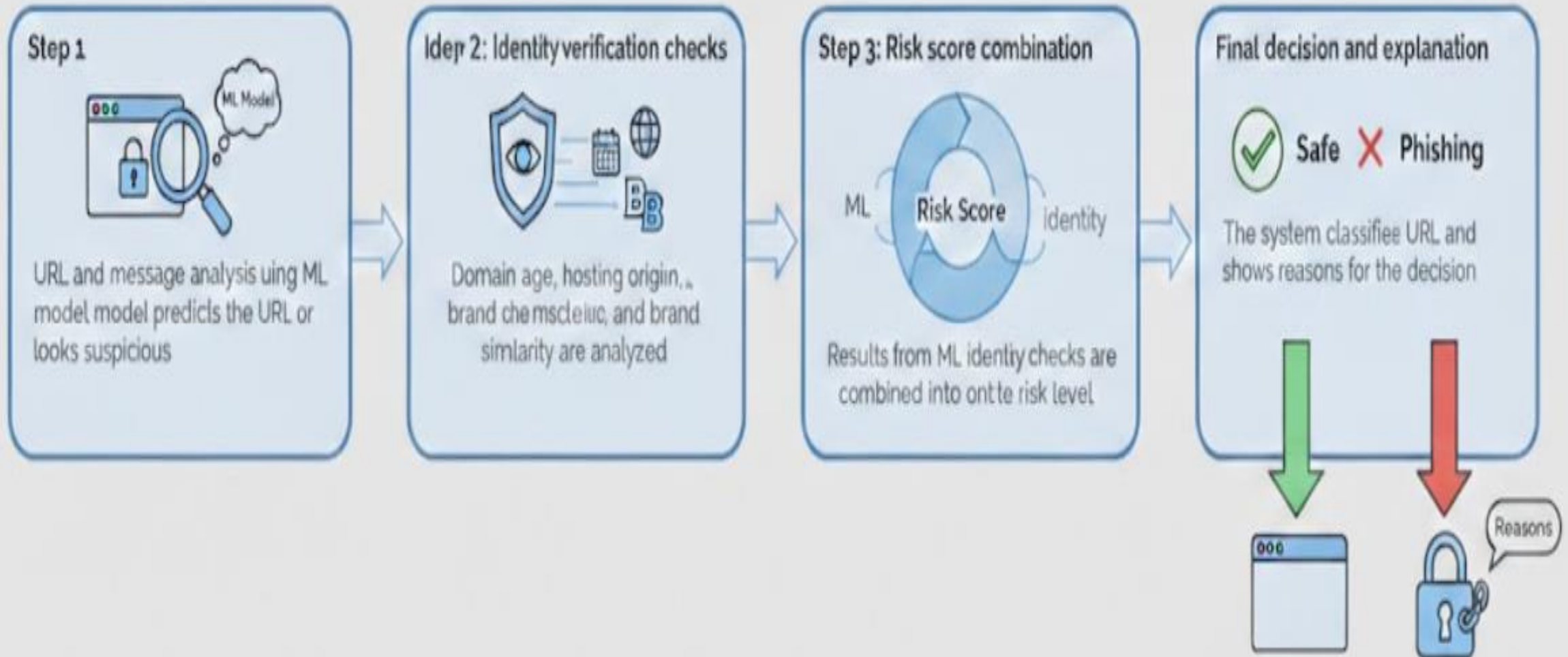
Verify whether a link truly belongs to the brand it claims to represent using domain, hosting, and brand-matching checks.

MsgGuard Features



- **AI-powered phishing detection** — ML model classifies suspicious messages and URLs
- **Message content analysis** — Detects risky language and urgent messages
- **Identity-based URL verification** — Checks WHOIS, hosting, and brand similarity
- **Zero-day phishing detection** — Identifies new attacks beyond blacklists
- **Brand impersonation detection** — Finds fake domains imitating real brands
- **Privacy-friendly design** — No storage of personal user data
- **Explainable alerts** — Shows reasons behind each warning

Phishing Detection Algorithm



Success Criteria

- **Correct detection with suspicious identity indicators** — URLs with new domains, brand mismatch, or unusual hosting are flagged in at least **85–90%** of cases
- **Identity checks improve detection accuracy** — Adding identity checks improves results by at least **5% over ML-only detection**
- **Clear explanation for each alert** — Every phishing decision includes at least one concrete reason (e.g., new domain, brand mismatch)

Challenges and Our Solution

Challenge	How MsgGuard Solves It
 Evolving Phishing Techniques	 Uses AI-based analysis to detect new and changing phishing patterns beyond static rules.
 Brand Impersonation	 Analyzes message language and verifies brand–domain similarity to identify fake identities.
 Zero-Day Attacks	 Applies identity-based URL verification (domain age, hosting origin) instead of relying on blacklists.
 Message-Only Phishing	 Detects phishing attempts based on message content, even when no link is included.

Dataset: Phishing & Legitimate URLs (Majestic Million)

Dataset Description

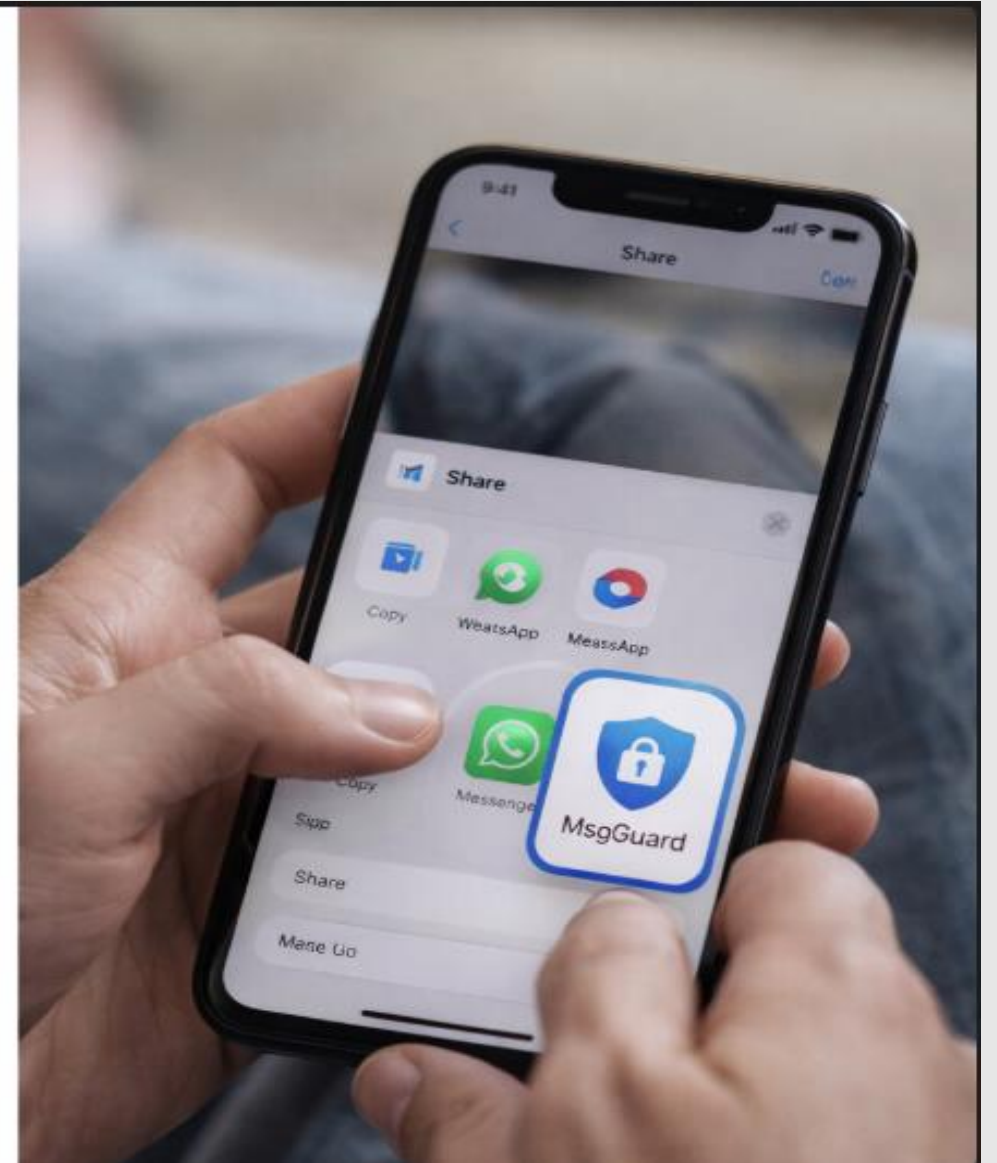
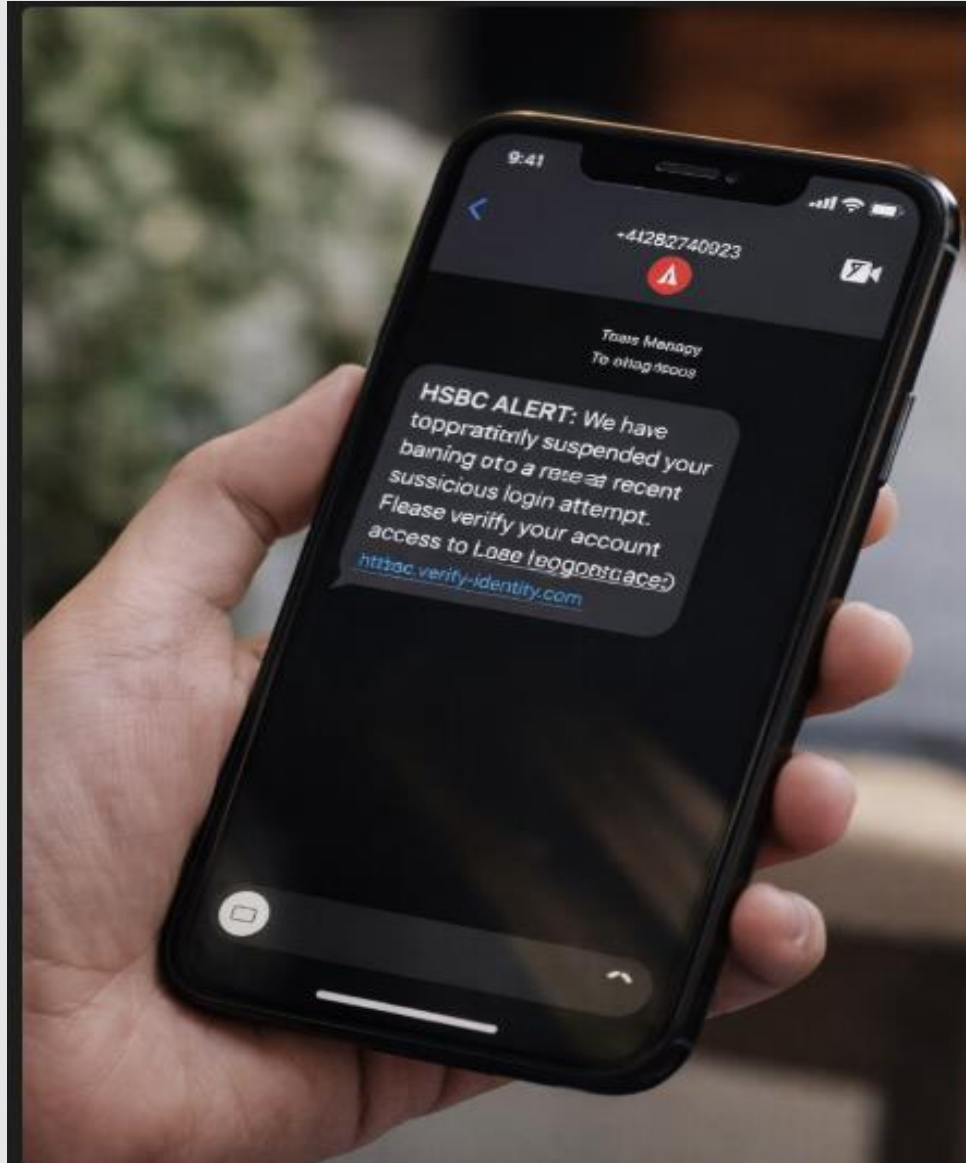
- Contains both **phishing** and **legitimate URLs**
- Collected from **public security sources**
- Includes many **brand impersonation websites**

The Dataset Is Used

- To **train and evaluate** the CNN-based URL model
- To learn **structural URL patterns** common in phishing
- To classify links as **legitimate or phishing**

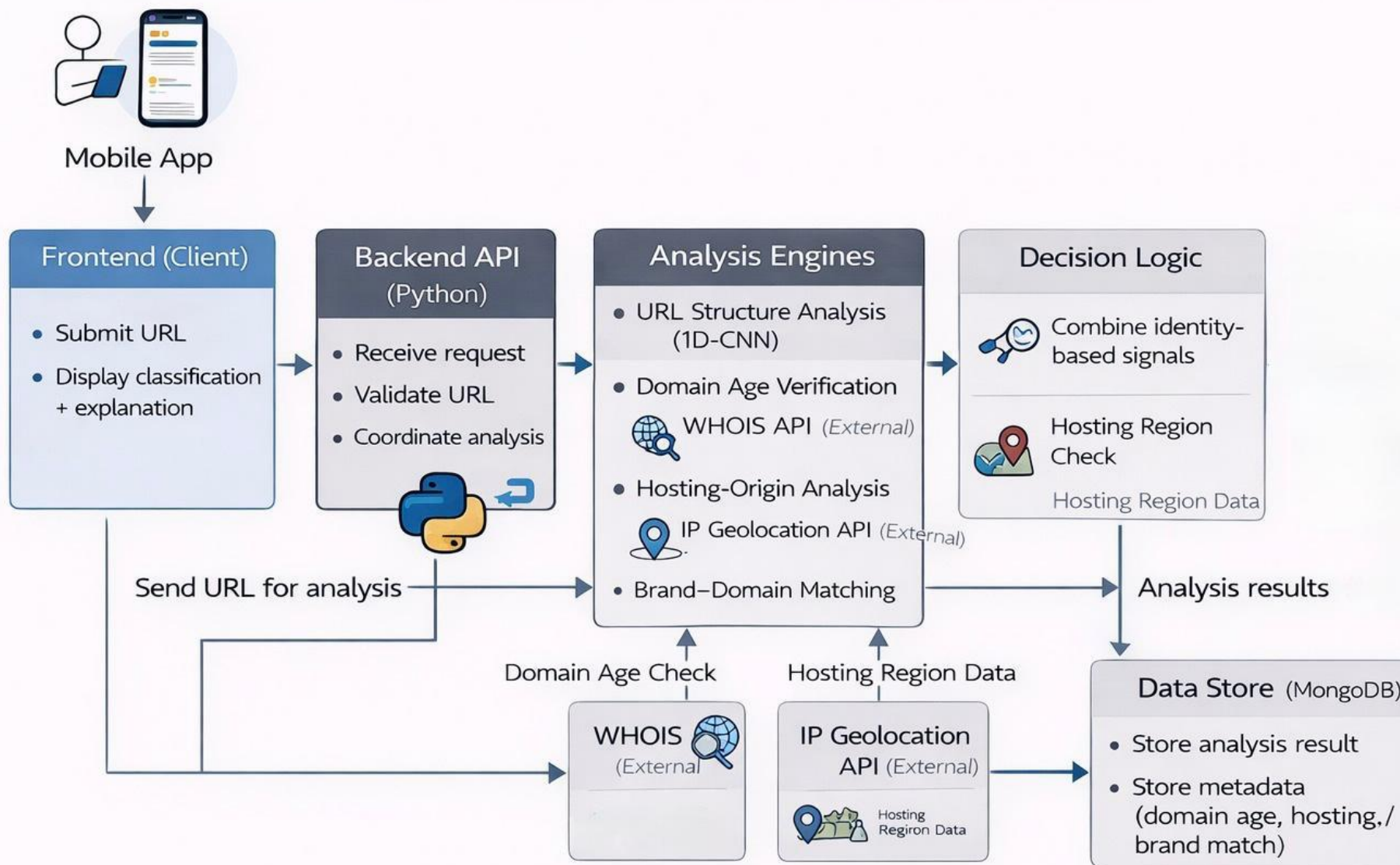


prototype

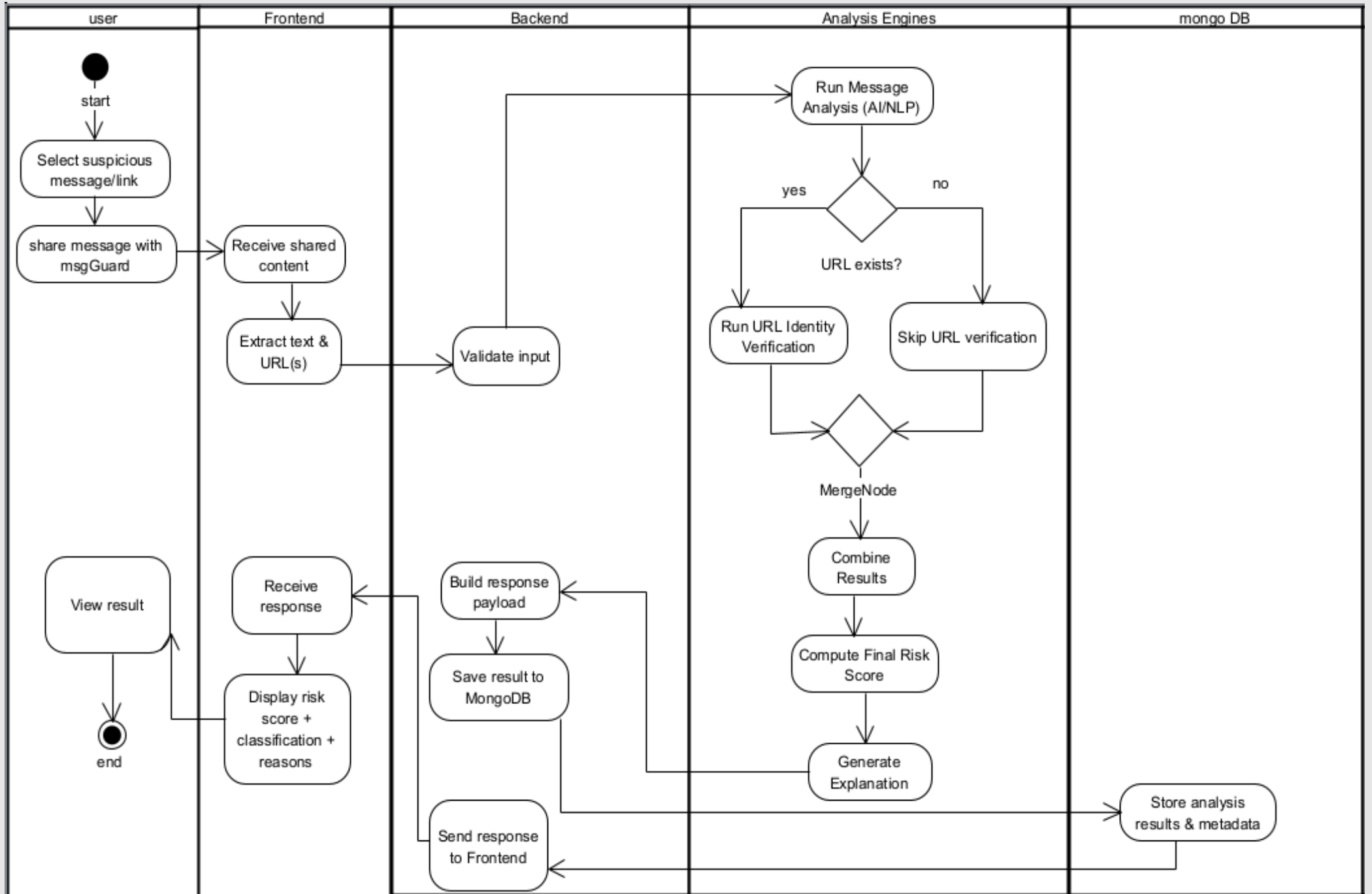




MsgGuard – System Architecture



Activity Diagram



Test Plan

Test Case	Scenario	What's Checked	Expected Result
TC1	Real Brand Website	Checks if the website is valid and the domain is old and trustworthy.	Classified as Legitimate
TC2	Newly Registered Website	Checks if the domain is very new and might be risky.	Classified as Suspicious or Phishing
TC3	Fake Brand Website	Checks if the website URL is trying to look like <u>a known brand's</u>	Classified as Phishing
TC4	Suspicious Hosting Location	Checks if the website is hosted in an unusual place.	Risk indicated in decision.

Verification and Evaluation

- The system is verified through end-to-end testing of message and link analysis.
- Multiple scenarios are evaluated, including phishing attempts, legitimate content, and zero-day domains.
- Results show reliable phishing detection with low false positives and false negatives.



Conclusion

- **Strong identity signals improve detection** — Website identity information adds important context beyond AI alone
- **Modules work better together** — Combining ML, domain checks, and brand matching improves decision reliability
- **Explainability supports user trust** — Clear reasons help users understand and react to threats
- **Approach fits real-time protection** — URL-level analysis enables fast and lightweight security checks

